

LGAI Group Assignment

Course: LLMs and Generative AI (LGAI), Indian School of Business

Group: Group-20

Part A: Four scenarios for each computer-vision task

For every task below, four scenarios are listed. Each scenario gives the business domain, the input data, the expected output, a company that has deployed it, and a reference.

1. Image Classification

Assigning a single label (or set of labels) to a whole image.

#	Domain	Input data	Expected output	Deployed by
1	Healthcare (Radiology)	Chest X-ray images	Label: normal, pneumonia or COVID-19	Qure.ai (qXR)
2	Agriculture	Smartphone photos of crop leaves	Crop-disease category (healthy, blight, rust and so on)	Plantix (PEAT GmbH)
3	E-commerce	Product catalogue images	Product category for catalogue tagging	Amazon
4	Manufacturing (QC)	Camera images of parts on a line	Defective vs non-defective	Landing AI (LandingLens)

2. Image Similarity Computation

Measuring how similar two images are using embeddings and a distance metric.

#	Domain	Input data	Expected output	Deployed by
1	E-commerce	A photo a user uploads plus the catalogue	Visually similar products (shop the look)	Pinterest (Pinterest Lens)
2	Security and identity	Two face images	Same-person similarity score for verification	Apple (Face ID)
3	Media and IP protection	Uploaded video frames vs a reference set	Near-duplicate or copyright match score	YouTube (Content ID)
4	Real estate	Listing photos	Find similar homes by interior style	Zillow

3. Object Detection

Locating and classifying many objects with bounding boxes.

#	Domain	Input data	Expected output	Deployed by
1	Autonomous driving	Vehicle camera stream	Boxes for cars, pedestrians and traffic signs	Tesla (Autopilot)
2	Retail	Shelf images	A box per product for planogram compliance	Trax Retail
3	Transportation (Rail)	Railway traffic dataset	Boxes for spanner, helmet, pedestrian, train and track	China Railway Bureau
4	Surveillance	CCTV footage	Boxes for persons, weapons and bags	Avigilon (Motorola)

4. Image Segmentation

Pixel-level labelling (semantic, instance or panoptic).

#	Domain	Input data	Expected output	Deployed by
1	Healthcare (Pathology)	H&E histopathology slides	Per-pixel nucleus or tumour mask	PathAI
2	Mobile photography	Phone portrait photo	Foreground and background mask (Portrait mode)	Google (Pixel)
3	Autonomous driving	Street-scene images	Per-pixel road, lane and sidewalk mask	Mobileye
4	Satellite and GIS	Satellite imagery	Building-footprint or land-cover masks	Microsoft (Building Footprints)

5. Visual Question Answering (VQA)

Answering a natural-language question about an image.

#	Domain	Input data	Expected output	Deployed by
1	Accessibility	Photo taken by a blind user plus a question	Spoken answer such as what is on this label	Microsoft (Seeing AI)
2	E-commerce support	Product image plus a customer question	Answer about colour, size or material	Amazon (Rufus)
3	Healthcare	Medical image	Answer about	Google (Med-

		plus a clinician question	findings	PaLM M)
4	Insurance	Damage photo plus an adjuster question	Answer about damage severity or part	Lemonade and Tractable

6. Action Recognition

Classifying the action or activity happening in a video clip.

#	Domain	Input data	Expected output	Deployed by
1	Fitness	Webcam workout video	Exercise type and rep counting	Tempo and Peloton Guide
2	Elderly care	Room camera video	Detect a fall vs normal activity	SafelyYou
3	Sports analytics	Match footage	Recognise pass, shot or tackle events	Stats Perform and Hawk-Eye
4	Manufacturing safety	Factory-floor CCTV	Detect unsafe actions such as no gloves	Intenseye

7. Multi-Object Tracking (MOT)

Detecting objects and keeping their identity across video frames.

#	Domain	Input data	Expected output	Deployed by
1	Smart cities and traffic	Intersection camera video	Per-vehicle trajectories and counts	NVIDIA (Metropolis)
2	Retail analytics	In-store camera video	Shopper paths and dwell time	Standard AI and Amazon Go
3	Sports broadcasting	Game footage	Track each player across the field	Second Spectrum (NBA)
4	Autonomous driving	LiDAR and camera stream	Track surrounding agents over time	Waymo

8. Image and Video Generation

Creating new images or videos from noise, text or a conditioning image.

#	Domain	Input data	Expected output	Deployed by
1	Marketing and design	Text prompt	Generated marketing image	OpenAI (DALL-E 3)
2	Gaming and film	Text or image prompt	Generated video clip	Runway (Gen-3)
3	E-commerce	Product plus scene prompt	Product placed in a lifestyle scene	Meta (AI ad backgrounds)

4	Fashion	Garment plus a person image	Virtual try-on image	Google (Try On in Search)
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Part B: Classification

Q1: Pretrained ResNet18, DenseNet121 and VGG19, 3 epochs

The pretrained backbones are used as fixed feature extractors. The backbone is frozen and only a new 15-way linear head is trained for 3 epochs.

Overall test accuracy:

Model	Overall accuracy	Mean precision	Mean recall
ResNet18	0.927	0.928	0.919
DenseNet121	0.976	0.968	0.968
VGG19	0.951	0.952	0.948

Per-class precision and recall tables for all three models are printed in q1.ipynb. DenseNet121 is the strongest. The lowest per-class scores for every model are on the visually similar pair crocodile and crocodile_head.

Q2: Fine-tuning VGG19, 3 epochs

Here the entire VGG19 network is fine-tuned end to end. All layers are trainable, with SGD at a learning rate of 1e-3. After 3 epochs:

- Overall test accuracy is 0.956 (macro precision 0.957, macro recall 0.953).
- Fully fine-tuned VGG19 slightly beats the frozen-backbone VGG19 from Q1 (0.951), with perfect precision and recall on dollar_bill, emu, pizza and windsor_chair.
- The hard pair crocodile and crocodile_head remains the main source of error.

Q3: Zero-shot CLIP-ViT-B/16

CLIP classifies each test image by comparing its embedding to the text embeddings of the prompts a photo of a <class>. It is not trained on the dataset.

- Overall zero-shot accuracy is 0.985 (macro precision 0.982, macro recall 0.983), the best of all Part B approaches.
- Eleven of the 15 classes reach precision and recall of 1.000.
- Only crocodile, crocodile_head, gramophone and nautilus drop slightly below 1.0.

Part B analysis: A large vision-language model (CLIP) trained on web-scale data beats both feature extraction and full fine-tuning of conventional CNNs without seeing a single training image, because these 15 Caltech-style categories map cleanly to everyday language concepts. Among the trained CNNs, the deeper backbone (DenseNet121) and full fine-tuning (VGG19) both help. Fine-grained, visually similar classes are the bottleneck for every method.

Part C: Segmentation

Dataset: 84 H&E pathology images at 2000 by 2000 RGB with binary nucleus masks (nucleus = 255). 60 images for training and 24 for testing. A UNet with a pretrained ResNet-18 encoder is fine-tuned for 15 epochs at 256 by 256 with a combined BCE plus Dice loss.

Nucleus-class, pixel-wise metrics on the 24 test images (threshold 0.5):

Metric	Value
Precision	0.669
Recall	0.867
F1 (Dice)	0.755
IoU	0.606

Analysis: The ImageNet-pretrained encoder lets the UNet locate nuclei well from only 60 images. Recall (0.867) is higher than precision (0.669): the model finds most nuclei but over-segments slightly, and down-sampling from 2000 by 2000 to 256 by 256 merges the thin borders between touching nuclei. The combined BCE plus Dice loss is important to counter the strong background to foreground imbalance. A qualitative image, ground truth and prediction triplet is shown in the notebook.

Part D: Object Detection

Dataset: 128 images with Pascal-VOC XML annotations. The classes obj1 to obj4 are mapped to Water Bottle, Milk Bottle, Tetra Pack and Can. The first 100 images are used for training and the last 28 for testing. A COCO-pretrained Faster R-CNN (ResNet-50 FPN) is fine-tuned for 10 epochs, and the box head is replaced for 5 outputs (4 objects plus background).

Per-class precision and recall at IoU of at least 0.5 (score threshold 0.5):

Class	TP	FP	FN	Precision	Recall	F1
Water Bottle	12	0	0	1.000	1.000	1.000
Milk Bottle	12	0	0	1.000	1.000	1.000
Tetra Pack	12	0	0	1.000	1.000	1.000
Can	14	0	0	1.000	1.000	1.000
Mean				1.000	1.000	1.000

Training loss fell smoothly from 0.553 at epoch 1 to 0.056 at epoch 10.

Analysis: Starting from a COCO-pretrained detector, the model adapts very quickly because the backbone already recognises generic bottle, box and can shapes. After 10 epochs it detects every test object correctly, with perfect precision and recall at IoU 0.5. The dataset is small and the four product classes are visually distinct and clearly imaged, so the fine-tuned detector generalises well to the 28 held-out images. Evaluation uses greedy IoU-0.5 matching, taking the highest-confidence predictions first. Unmatched predictions would count as false positives and unmatched ground-truth boxes as false negatives. Here there were none.

